Fernando Contreras
Assignment 7.2: Unit Testing Assignment
May 1, 2026

a) Write a function that accepts two parameters: a city name and a country name. The function should return a single string of the form City, Country, such as Santiago, Chile. Store the function in a file named city_functions.py. In the same file, call the function at least three times using a City, Country values. Execute city_functions.py and take a screenshot of the result. Paste that screenshot into your Word document.

city_functions.py U

csd-325 > module-7 > city_functions.py > ...

```python
"""
city_functions.py
Author: Fernando Contreras
Assignment: CSD-325 Module 7
Description: Function that accepts a city and country name and returns
             a formatted string in the form City, Country.
"""


def city_country(city, country):
    """Return a formatted City, Country string.

    Parameters:
        city (str): Name of the city.
        country (str): Name of the country.

    Returns:
        str: Formatted string, e.g. 'Santiago, Chile'
    """
    return f"{city.title()}, {country.title()}"
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh

```
/usr/bin/python3 /Users/fernandocontreras/Documents/Bellevue/csd/csd-325/module-7/city_functio
.py
(base) fernandocontreras@Fernandos-MacBook-Pro csd % /usr/bin/python3 /Users/fernandocontreras
ocuments/Bellevue/csd/csd-325/module-7/city_functions.py
Santiago, Chile
Tokyo, Japan
Paris, France
(base) fernandocontreras@Fernandos-MacBook-Pro csd %
```

b) Create a file called test_cities.py that tests the function you just wrote (remember that you need to import unittest and the function you want to test). Write a method called test_city_country() to verify that calling your function with values such as Santiago and Chile results in the correct string. Run test_cities.py, and make sure test_city_country() passes. When it passes, take a screenshot of the result and paste in into your Word document.

city_functions.py U test_cities.py U

~/Documents/Bellevue/csd/csd-325/module-7/city_functions.py • Untracked

```python

import unittest
from city_functions import city_country


class TestCityCountry(unittest.TestCase):
    """Tests for the city_country function."""

    def test_city_country(self):
        """Test city and country only."""
        result = city_country("santiago", "chile")
        self.assertEqual(result, "Santiago, Chile")


if __name__ == "__main__":
    unittest.main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL zsh - module-7

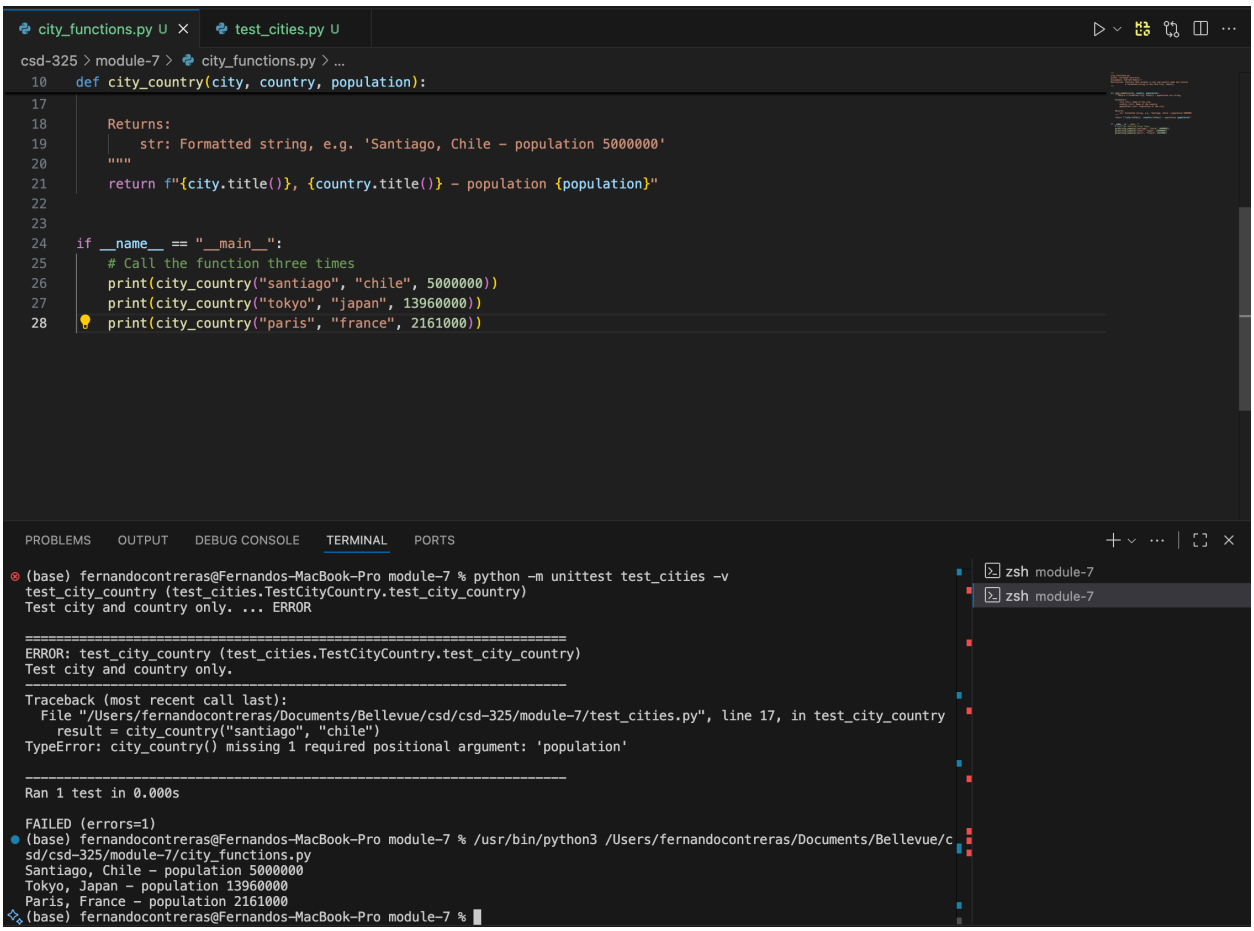
```
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % python -m unittest test_cities -v
Santiago, Chile
Tokyo, Japan
Paris, France
test_city_country (test_cities.TestCityCountry.test_city_country)
Test city and country only. ... ok

----------------------------------------------------------------------
Ran 1 test in 0.000s

OK
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 %
```

c) Modify your city_country function in city_functions.py so it requires a third parameter, population. It should now return a single string of the form City, Country - population xxx, such as Santiago, Chile - population 5000000.

d) Run test_cities.py again. It should fail. Take a screenshot of the result and paste into your Word document.



e) Now modify your city_country function in city_functions.py so that the population parameter is optional.

f) Run test_cities.py again. It should pass. Take a screenshot of the result and paste into your Word document.

city_functions.py U × test_cities.py U

csd-325 > module-7 > city_functions.py > ...

```python
def city_country(city, country, population=None):
    Returns:
        str: Formatted string, e.g. 'Santiago, Chile - population 5000000'
    """
    result = f"{city.title()}, {country.title()}"
    if population:
        result += f" - population {population}"
    return result


if __name__ == "__main__":
    # Call the function three times
    print(city_country("santiago", "chile", 5000000))
    print(city_country("tokyo", "japan", 13960000))
    print(city_country("paris", "france", 2161000))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % python -m unittest test_cities -v
test_city_country (test_cities.TestCityCountry.test_city_country)
Test city and country only. ... ok

----------------------------------------------------------------------
Ran 1 test in 0.000s

OK
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % /usr/bin/python3 /Users/fernandocontr
sd/csd-325/module-7/city_functions.py
Santiago, Chile - population 5000000
Tokyo, Japan - population 13960000
Paris, France - population 2161000
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 %
```

g) Modify your city_country function in city_functions.py so it requires a fourth parameter, language. It should now return a single string of the form City, Country - population xxx, Language, such as Santiago, Chile - population 5000000, Spanish.

h) Run test_cities.py again. It should fail. Take a screenshot of the result and paste into your Word document.

city_functions.py U × test_cities.py U

csd-325 > module-7 > city_functions.py > ...

```python
def city_country(city, country, population=None, language=None):
        language (str): Primary language spoken.

    Returns:
        str: Formatted string, e.g. 'Santiago, Chile - population 5000000, Spanish'
    """
    result = f"{city.title()}, {country.title()}"
    if population:
        result += f" - population {population}"
    result += f", {language.title()}"
    return result
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % python -m unittest test_cities -v
test_city_country (test_cities.TestCityCountry.test_city_country)
Test city and country only. ... ERROR

======================================================================
ERROR: test_city_country (test_cities.TestCityCountry.test_city_country)
Test city and country only.
----------------------------------------------------------------------
Traceback (most recent call last):
  File "/Users/fernandocontreras/Documents/Bellevue/csd/csd-325/module-7/test_cities.py", line 17, in test_city_country
    result = city_country("santiago", "chile")
  File "/Users/fernandocontreras/Documents/Bellevue/csd/csd-325/module-7/city_functions.py", line 25, in city_country
    result += f", {language.title()}"
                   ^^^^^^^^^^^^^^
AttributeError: 'NoneType' object has no attribute 'title'

----------------------------------------------------------------------
Ran 1 test in 0.001s

FAILED (errors=1)
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 %
```

i) Now modify your city_country function in city_functions.py so that the language argument is optional.

j) Run test_cities.py again. It should pass. Take a screenshot of the result and paste into your Word document.

city_functions.py U × test_cities.py U

csd-325 > module-7 > city_functions.py > ...

```python
def city_country(city, country, population=None, language=None):
        str: Formatted string, e.g. 'Santiago, Chile - population 5000000, Spanish'
    """
    result = f"{city.title()}, {country.title()}"
    if population:
        result += f" - population {population}"
    if language:
        result += f", {language.title()}"
    return result


if __name__ == "__main__":
    # Call the function three times
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % python -m unittest test_cities -v
  File "/Users/fernandocontreras/Documents/Bellevue/csd/csd-325/module-7/test_cities.py", line
    result = city_country("santiago", "chile")
  File "/Users/fernandocontreras/Documents/Bellevue/csd/csd-325/module-7/city_functions.py", li
    result += f", {language.title()}"
                   ^^^^^^^^^^^^^^
AttributeError: 'NoneType' object has no attribute 'title'

----------------------------------------------------------------------
Ran 1 test in 0.001s

FAILED (errors=1)
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % python -m unittest test_cities -v
test_city_country (test_cities.TestCityCountry.test_city_country)
Test city and country only. ... ok

----------------------------------------------------------------------
Ran 1 test in 0.000s

OK
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 %
```

k) Run city_functions.py. Call the function at least three times using a City, Country for one, City, Country, Population for the next and City, Country, Population, Language for the last. Execute city_functions.py and take a screenshot of the result. Paste that screenshot into your Word document.

city_functions.py U × test_cities.py U

csd-325 > module-7 > city_functions.py > city_country

```python
Assignment: CSD-325 Module 7
Description: Function that accepts a city and country name and returns
             a formatted string in the form City, Country.
"""


def city_country(city, country, population=None, language=None):
    """Return a formatted string with city, country, optional population and langua

    Parameters:
        city (str): Name of the city.
        country (str): Name of the country.
        population (int, optional): Population of the city.
        language (str, optional): Primary language spoken.

    Returns:
        str: Formatted string, e.g. 'Santiago, Chile - population 5000000, Spanish'
    """
    result = f"{city.title()}, {country.title()}"
    if population:
        result += f" - population {population}"
    if language:
        result += f", {language.title()}"
    return result
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % python -m unittest test_cities -v

FAILED (errors=1)
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % python -m unittest test_cities -v
test_city_country (test_cities.TestCityCountry.test_city_country)
Test city and country only. ... ok

----------------------------------------------------------------------
Ran 1 test in 0.000s

OK
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 % /usr/bin/python3 /Users/fernandoco
sd/csd-325/module-7/city_functions.py
Santiago, Chile - population 5000000, Spanish
Tokyo, Japan - population 13960000, Japanese
Paris, France - population 2161000, French
(base) fernandocontreras@Fernandos-MacBook-Pro module-7 %
```